

Algorithms Problem Solving

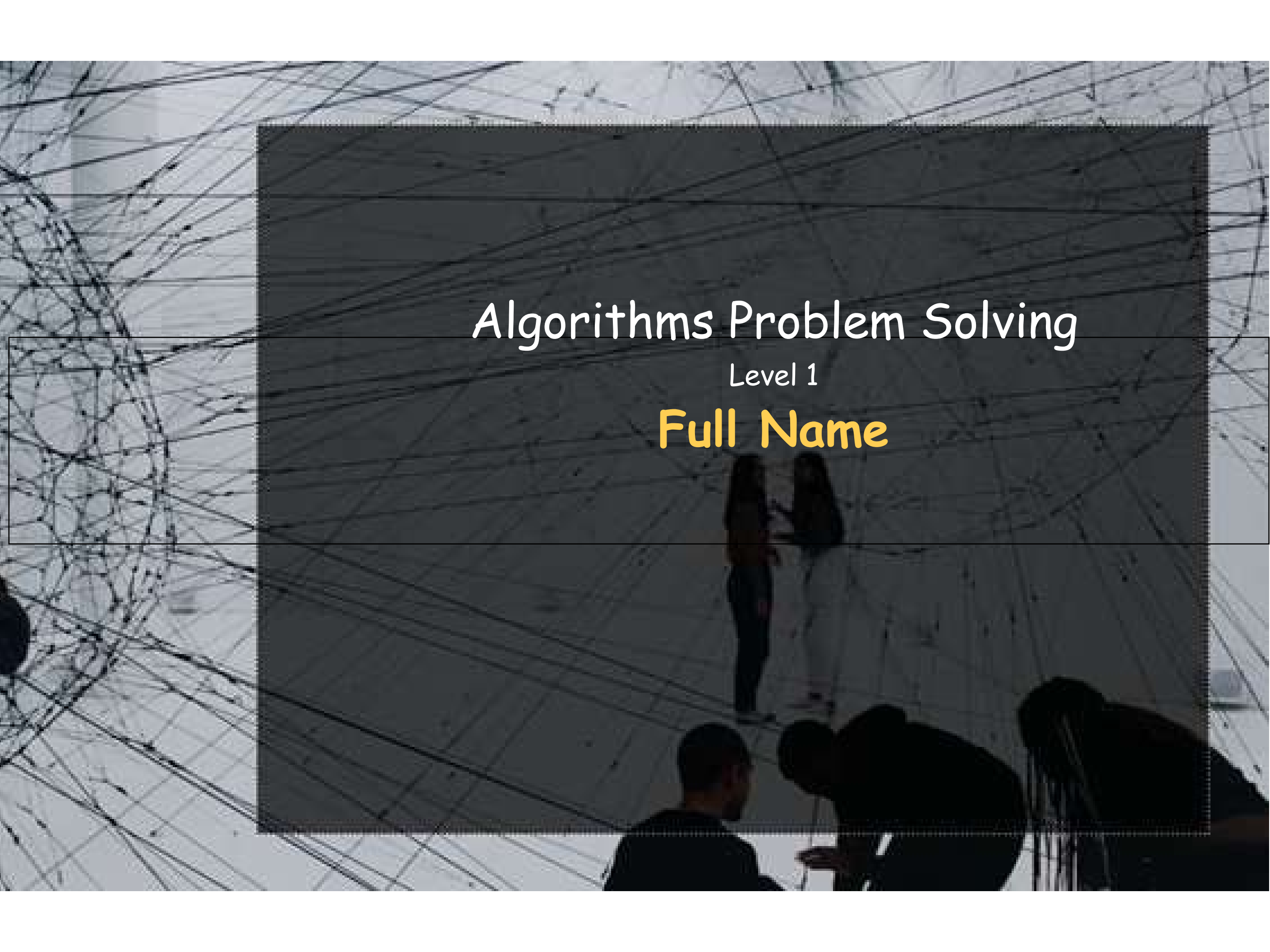
Level 1

Full Name

Algorithms Problem Solving

Level 1

Full Name



Definition

- ◆ This program reads the user's first name and last name separately and then displays the full name.
- ◆ The program uses a structure, functions, string concatenation, and a Boolean parameter.
- ◆ It can display the full name in both normal and reversed order.

Explanation

- ◆ The struct `stName` groups the first and last names:
 - `firstName` stores the user's first name.
 - `lastName` stores the user's last name.
- ◆ `readName()` reads the first name and last name using `getline()` and returns an `stName` object.
- ◆ `getFullName()` combines the first name and last name into one string using the `+` operator.
- ◆ `getFullNameReversed()` can return the name in normal or reversed order depending on the reversed Boolean parameter.
- ◆ If `reversed` is `true`, the last name is displayed first.
- ◆ If `reversed` is `false`, the first name is displayed first.
- ◆ `printFullName()` prints both versions of the full name.
- ◆ In `main()`, the functions are combined:
`printFullName(readName());`

Steps in main()

1. Create the stName structure.
2. Read the first name.
3. Read the last name.
4. Return the collected names.
5. Pass the structure to printFullName().
6. Combine the first and last names using getFullName().
7. Use getFullNameReversed() to display the name in reversed order.
8. Print both versions of the full name.

💡 Example

◆ If the user enters:

First Name: Sara

Last Name: Alterkawi

◆ `getFullName()` returns:

Sara Alterkawi

◆ `getFullNameReversed()` with `reversed = true` returns:

Alterkawi Sara

➡ Result

The program displays:

Your Full Name is: Sara Alterkawi.

Your Full Name is: Alterkawi Sara

Note

- ◆ `getline()` is used to read the names, so names containing spaces can be read correctly.
- ◆ The expression `name.firstName + " " + name.lastName` concatenates the two names with a space between them.
- ◆ The `reversed` parameter controls the order of the names.

⊘ Warning

- ◆ The parameter reversed must be set correctly depending on the desired order.
- ◆ In this program, printFullName() passes true, so the second output is displayed in reversed order.



Comparison

▨ Normal Full Name

- ✓ First name comes first.
- ✓ Uses firstName + " " + lastName.
- ✓ Example: Sara Alterkawi.

▤ Reversed Full Name

- ✓ Last name comes first.
- ✓ Uses lastName + " " + firstName.
- ✓ Example: Alterkawi Sara.

✦✦ Summary

- ✓ A structure can group first and last names into one object.
- ✓ String concatenation can be used to create a full name.
- ✓ A Boolean parameter can control whether the name is displayed normally or in reversed order.

Thank you 😊